

CSA0508 DATABASE MANAGEMENT SYSTEM

A COMPARATIVE STUDY

File-Based Systems vs. DBMS

Real-world lessons from Banking and Healthcare — how the shift from isolated files to centralized databases reshaped two industries.

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What We'll Cover

1

Foundations

Defining file-based systems and DBMS from first principles

2

Head-to-Head

A structured comparison across integrity, security, and scale

3

Banking Case Study

State Bank of India's move to Core Banking Solutions

4

Healthcare Case Study

Kaiser Permanente's HealthConnect EHR transformation

5

Deep-Dive Analysis

Redundancy, security, scalability, and cost, compared

6

Takeaways

Lessons for any organization planning a similar migration

What Is a File-Based System?

A file-based system stores data in separate, independent files managed directly by application programs, rather than through a shared database engine. Each application defines and owns its own file format.



Data Redundancy

The same customer or patient details are re-entered and stored in multiple, disconnected files.



No Central Control

Every application program handles its own file structure, access logic, and validation.



Rigid Structure

Adding a new field or report often means rewriting application code, not just the data.

isolated file

Accounts.dat

isolated file

Customers.dat

isolated file

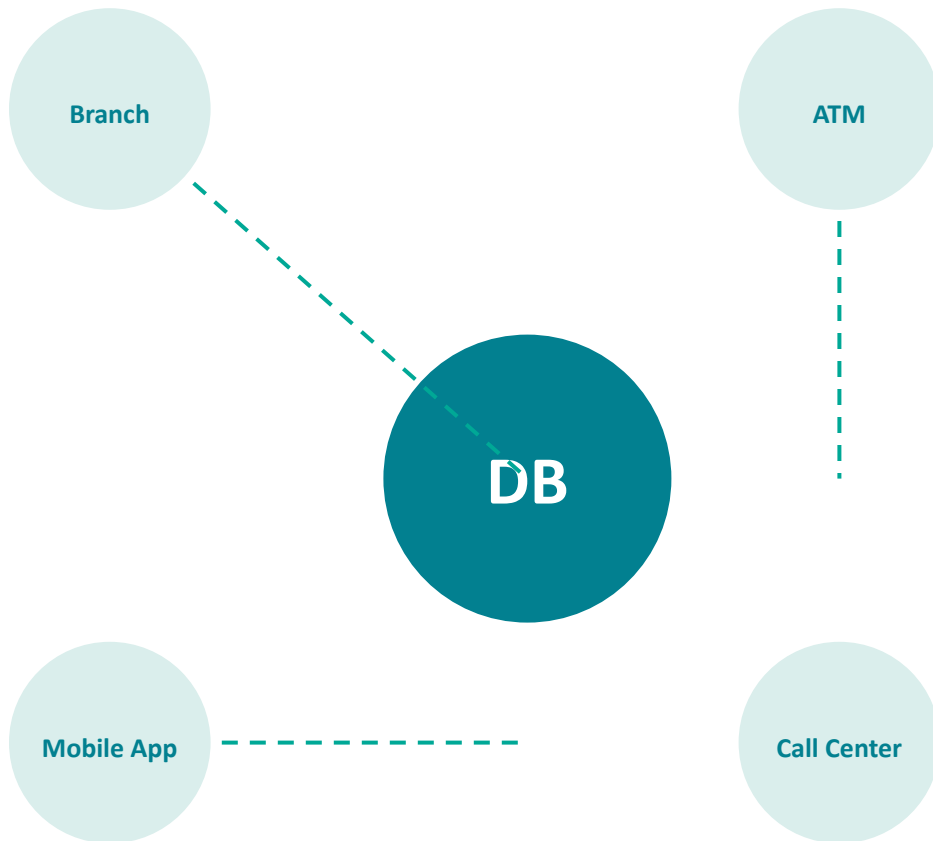
Loans.dat

isolated file

Branch_Ledger.dat

No shared structure — each file speaks its own format

What Is a DBMS?



A Database Management System is centralized software that stores, organizes, and controls access to data for many applications at once — enforcing one consistent, shared structure and set of rules.

1

Single Source of Truth

One consistent record per customer or patient, shared across every access point.

#

Built-in Controls

Integrity constraints, transactions, and role-based security are enforced by the DBMS itself.

+

Concurrent, Scalable Access

Thousands of users and branches read and write safely at the same time.

File-Based Systems vs. DBMS

Aspect	File-Based System	DBMS
Data Redundancy	High — duplicated across files	Minimized via normalization
Data Integrity	Enforced manually by programs	Enforced by constraints & rules
Security	Basic, file-level, inconsistent	Fine-grained, role-based access
Concurrent Access	Prone to conflicts, limited	Managed via transactions & locks
Backup & Recovery	Manual, error-prone	Automated, transaction-safe
Scalability	Difficult across locations	Built for enterprise-wide scale

Life Before Core Banking

State Bank of India, pre-2003

Through the 1990s, SBI computerized individual branches, but each branch ran its own standalone, file-based system. Branch automation was completed by 2004 — yet the branches still could not talk to one another.

A

No “Anywhere Banking”

A customer's account existed only at the branch where it was opened — service elsewhere was impossible.

B

No Single Customer View

The same customer's deposits, loans, and cards lived in separate, disconnected branch files.

C

Manual Reconciliation

End-of-day tallying and inter-branch settlement relied on manual, paper-backed processes.

D

A National-Scale Bottleneck

By 2003, only about 9% of Indian bank branches had even reached centralized systems.

The Core Banking Solution

A centralized, DBMS-driven transformation

SBI migrated from isolated branch files to TCS BaNCS, a Core Banking Solution built on a centralized database. Every branch, ATM, and online channel now reads and writes the same real-time customer record.

100M+

accounts unified on one centralized database

9% → 50%

of Indian bank branches on core banking,
2003–2010

24/7

real-time transactions across every channel

Result: a single customer view, disaster-recovery-backed uptime, and the database backbone that today powers NEFT, RTGS, and UPI at national scale.

Fragmented Patient Records

Kaiser Permanente, pre-HealthConnect

Kaiser Permanente operated across 8 largely independent regions. Each ran its own information systems, sharing only limited data elements — the classic symptoms of disconnected, file-based record-keeping.

A

Regional Silos

Eight regions, eight sets of systems — each with its own patient data trapped locally.

B

Costly Customization

Disparate IT systems required constant, expensive, one-off maintenance to keep running.

C

Limited Data Sharing

A specialist or ER visit elsewhere often meant re-collecting history from scratch.

D

Care Coordination Risk

Incomplete records made it harder to catch drug interactions or duplicate tests.

KP HealthConnect

The largest private-sector electronic health record

Kaiser Permanente built a single, database-backed spanning every region. One record now follows the patient across primary care, specialists, pharmacy, and hospital visits.

431 + 36

medical offices and hospitals on one shared record

8.6M+

people connected to physicians, nurses & pharmacists

723K

patients in a single EHR-derived research registry

Result: a near-paperless environment with built-in clinical decision support, medication bar-coding, and data warehoused for population-health research.

Cost, Maintenance & Recovery

F

File-Based Approach

- Each local system needs its own upkeep, patches, and support
- Kaiser's regional systems required costly, one-off customization
- Backup and recovery is manual and inconsistent across sites

D

DBMS Approach

- One platform to maintain, upgrade, and secure centrally
- SBI invested in dedicated data centers with disaster recovery
- Automated, transaction-safe backups protect the single source of truth

Case tie-in: consolidation raised upfront investment for both organizations, but lowered the long-run cost of keeping many disconnected systems alive.

CONCLUSION

From Isolated Files to a Shared Source of Truth

Whether it's a bank branch's ledger or a hospital's patient chart, file-based systems break down at scale. State Bank of India and Kaiser Permanente reached the same conclusion from opposite industries: centralizing data in a well-designed DBMS is what makes real-time, secure, and reliable service possible.

One record

replaces many disconnected copies

Built-in rules

replace manual, error-prone checks

Real-time scale

replaces branch-by-branch limits

Thank you — Questions & Discussion

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THANK YOU